

Adaptive Parity–Hall Reduction for Native Möbius Rows

A Single-Scalar Julia Lift and Activation-Boundary Frontier

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Abstract

A parity-aware reconstruction of a projective rough-prime route reduced the positivity of native Möbius rows to a global common-source problem, but proved that every fixed local parity reset fails. We continue that route in three steps. First, the unique conclusion-producing combination $\mathcal{R}_X = 5c_X(2) + 3c_X(3)$ collapses the two-row cone to one reciprocal state whose Mellin numerator has no zero in the relevant half-plane. Second, an adaptive Bonferroni theorem proves that variable even depth restores positivity in the complete homogeneous rough product, while a repeated-integral formula proves that every smooth-interior rough finite difference is nonnegative. Hence the remaining sign is supported only on histories crossing a row activation boundary. Third, we construct an exact positive 2×2 Julia lift in which cumulative rough-history parity acts by diagonal conjugation and the off-diagonal is precisely the 5:3 scalar. The positive trace retains the real zeta pole, so matrix positivity alone cannot feed Landau. The entire unresolved producer is localized to one globally owned activation-boundary functional, denoted GABPT. We prove GABPT implies the Riemann Hypothesis through one fixed Mellin transform. GABPT is not proved here; RH remains unproved.

Status. This is an unconditional reduction and interface repair, not a proof of RH. The remaining arithmetic theorem is stated explicitly as GABPT.

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1 The native rows and the unique scalar

For the canonical component row $Q_X(j)$, write

$$c_X(j) = \sum_{k \leq X/j} \frac{\mu(k)}{\sqrt{k}} Q_{X/k}(j).$$

The parity audit of the predecessor shows that cumulative rough history changes the sign of a terminal observation by $(-1)^{|h|}$; magnitudes and activations remain unchanged. The two fixed rows have an exact conclusion-producing combination.

Theorem 1.1 (Single-scalar collapse). *Put $\mathcal{R}_X = 5c_X(2) + 3c_X(3)$. Then*

$$\mathcal{R}_X = \sum_{n \leq X} \frac{a_*(n)}{\sqrt{n}} \log \frac{X}{n},$$

where

$$a_*(n) = 6\mathbf{1}_{n=1} - 6\mu(n) + 9\mathbf{1}_{2|n}\mu(n/2) - 3\mathbf{1}_{4|n}\mu(n/4).$$

The unsieved dictionary $q_* = 1 * a_*$ is

$$q_*(1) = 0, \quad q_*(2) = 15, \quad q_*(3) = 6, \quad q_*(4) = 3, \quad q_*(m) = 6 \quad (m \geq 5).$$

Proof. Insert the exact coefficient dictionaries of rows two and three and collect the Möbius shifts. Convolution with the constant-one sequence gives the displayed finite values; for $m \geq 5$ every boundary correction has entered and the constant tail is six. $\square$

Let $b = (\delta_1 - \delta_2) * \mu$ and

$$G(X) = \sum_{n \leq X} \frac{b(n)}{\sqrt{n}} \log \frac{X}{n}.$$

Direct convolution gives

$$a_* = 6\delta_1 - 3(2\delta_1 - \delta_2) * b,$$

hence

$$\mathcal{R}_X = 6 \log X - 3 \left(2G(X) - 2^{-1/2}G(X/2) \right). \tag{1}$$

This is the first simplification: the global parity-Hall problem has only one scalar output and one reciprocal state.

2 The fixed-depth obstruction and adaptive depth

The predecessor proved that every fixed even depth eventually makes the current Euler block negative, whereas every fixed odd depth leaves the recursive parity reversed. This rules out a fixed local reset, not a variable stopping rule.

Let $0 \leq r_i \leq r_{\max} < 1$, let e_k be the elementary symmetric sums, and put $z = \sum_i r_i$. The complete rough product is

$$P = \prod_i (1 - r_i) = \sum_{k \geq 0} (-1)^k e_k > 0.$$

Theorem 2.1 (Adaptive Bonferroni positivity). *For an even integer $L > z$, put*

$$S_{L-1} = \sum_{k=0}^{L-1} (-1)^k e_k.$$

If

$$L \log \frac{L}{ez} > \frac{z}{1 - r_{\max}},$$

then $S_{L-1} > 0$. For rough-prime weights $r_i = p_i^{-1/2}$, the smallest even integer with $L \geq 8(z+1)$ is sufficient.

Proof. Since $-\log(1-r) \leq r/(1-r_{\max})$,

$$P \geq \exp\left(-\frac{z}{1-r_{\max}}\right).$$

Maclaurin's inequality gives $e_k \leq z^k/k!$. A Chernoff bound for the exponential tail yields

$$\sum_{k \geq L} e_k \leq \sum_{k \geq L} \frac{z^k}{k!} \leq \left(\frac{ez}{L}\right)^L.$$

The assumed inequality makes the absolute omitted tail smaller than P , so $S_{L-1} = P - \sum_{k \geq L} (-1)^k e_k > 0$. For $r_{\max} \leq 67^{-1/2}$, the explicit linear choice has ample room. $\square$

The theorem explains why the fixed-depth asymptotic does not settle the global problem: the correct depth may grow with the accumulated rough hazard.

3 Smooth-interior positivity

Critical conjugation turns the logarithmic ramp into

$$\phi(u) = ue^{u/2} \mathbf{1}_{u \geq 0}.$$

Lemma 3.1 (Positive derivatives). *For $r \geq 1$ and $u > 0$,*

$$\phi^{(r)}(u) = e^{u/2} \left(\frac{u}{2^r} + \frac{r}{2^{r-1}} \right) > 0.$$

Proof. Differentiate $ue^{u/2}$ and induct on r . $\square$

Theorem 3.2 (Smooth-interior finite differences). *If $a_i \geq 0$ and $u \geq a_1 + \dots + a_r$, then*

$$\prod_{i=1}^r (I - T_{a_i}) \phi(u) = \int_0^{a_1} \dots \int_0^{a_r} \phi^{(r)}(u - v_1 - \dots - v_r) dv_1 \dots dv_r \geq 0.$$

Proof. Apply $f(u) - f(u-a) = \int_0^a f'(u-v) dv$ repeatedly. The activation assumption keeps the complete integration cube in the smooth region, and Lemma 3.1 supplies the sign. $\square$

Thus a rough history can contribute negatively only when its translation cube crosses an activation boundary. This localizes the producer sharply.

4 Exact activation-boundary localization

Fully expand the finite parity-labelled source at fixed endpoint X . Assign every history to the smooth-interior class if its complete translation cube stays within one active row cell; assign it to the boundary class otherwise. One-use ownership gives the exact decomposition

$$\mathcal{R}_X = \mathcal{I}_X + \mathcal{B}_X. \quad (2)$$

Theorem 3.2 gives $\mathcal{I}_X \geq 0$. The remaining statement is:

Definition 4.1 (GABPT). *The Global Activation-Boundary Parity Transport theorem is*

$$\mathcal{B}_X \geq -\mathcal{I}_X$$

for every sufficiently large X , with every cumulative history parity and source coefficient used exactly once.

GABPT is equivalent to eventual nonnegativity of $\mathcal{R}_X$. It is a global, variable-depth theorem. Separate leafwise transports cannot prove it, because odd histories reverse orientation and independent leaf budgets duplicate the same source reserve.

5 A positive parity-covariant Julia lift

Let $g_2(n) = v_2(n) + 1$ and

$$h_*(n) = 6g_2(n) + 3\mathbf{1}_{2|n}g_2(n/2).$$

Define

$$\mathcal{J}_*(n) = \begin{pmatrix} h_*(n) & a_*(n) \\ a_*(n) & h_*(n) \end{pmatrix}.$$

Theorem 5.1 (Julia positivity and parity conjugation). *For every n , $h_*(n) \geq |a_*(n)|$, so $\mathcal{J}_*(n) \succeq 0$. If $D = \text{diag}(1, -1)$, then for every rough history h ,*

$$D^{|h|} \mathcal{J}_*(n) D^{|h|} \succeq 0,$$

and its off-diagonal is $(-1)^{|h|} a_*(n)$.

Proof. For an odd squarefree core $d > 1$, the only nonzero dyadic levels have

$$(|a_*|, h_*) = (6, 6), (15, 15), (12, 24), (3, 33).$$

The unit core is checked directly, and nonsquarefree odd cores have zero scalar coefficient. Diagonal conjugation preserves the PSD cone and flips the off-diagonal once per parity swap. $\square$

The theorem repairs the parity interface exactly: one need not positively realize both a scalar datum and its negative. Both are off-diagonal faces of one positive matrix state.

6 Why the positive trace does not finish Landau

The Dirichlet series of g_2 is

$$\sum_{n \geq 1} \frac{g_2(n)}{n^z} = \frac{\zeta(z)}{1 - 2^{-z}}.$$

After excluding the nonphysical unit column, the Julia diagonal has symbol

$$H(z) = 3(2 + 2^{-z}) \frac{\zeta(z)}{1 - 2^{-z}} - 6.$$

It retains the pole at $z = 1$.

Firewall 6.1 (Source-blind Julia promotion). *The positive eigenchannels $H(z) \pm A(z)$ retain the same real pole. Their Mellin abscissa is therefore the shifted value $1/2$, not zero. PSD Julia gluing preserves parity but cannot, by its trace alone, provide the Landau continuation needed for RH. The trace-free off-diagonal scalar remains the conclusion-producing quantity.*

7 The annular reserve

The complete P_{61} scale-four annular theorem supplies strong local lower bounds for rows two and three. Its local certificate is useful, but its parity-blind root composition cannot be imported after the odd-history counterexample. In the present architecture an annular reserve may be used only as one globally owned budget against $\mathcal{B}_X$. It may not be copied among leaves, used in canonical orientation after odd histories, or reclassified as recursive source mass. Proving the required one-use domination is one possible route to GABPT.

8 The Mellin–Landau consumer

Theorem 8.1 (Exact scalar transform). *For $z = s + 1/2$, initially in the absolute half-plane,*

$$\int_1^\infty \mathcal{R}_X X^{-s-1} dX = \frac{6}{s^2} - \frac{3(1 - 2^{-z})(2 - 2^{-z})}{s^2 \zeta(z)}.$$

Proof. Integrate each logarithmic ramp using

$$\int_n^\infty \log(X/n) X^{-s-1} dX = \frac{n^{-s}}{s^2},$$

then insert the Dirichlet series of a_* . □

Theorem 8.2 (GABPT implies RH). *If GABPT holds, then the Riemann Hypothesis holds.*

Proof. GABPT makes $\mathcal{R}_X$ nonnegative for all sufficiently large X . Removing a finite initial interval changes its Mellin transform by an entire function. Landau’s theorem forces the tail transform to be holomorphic through $\Re s > 0$, since the displayed meromorphic expression has no positive-real singularity. A zero ρ of ζ with $\Re \rho > 1/2$ would create a pole at $s = \rho - 1/2$. The numerator cannot cancel it: its zeros require $2^{-\rho} = 1$ or 2 , which force $\Re \rho = 0$ or -1 . This is a contradiction. Functional symmetry gives RH. □

9 Hostile reconstruction checklist

Reject the route at the first occurrence of any of the following:

1. a rough history whose cumulative parity is recorded but not applied;
2. an activation-boundary history counted in both $\mathcal{I}_X$ and $\mathcal{B}_X$;
3. an annular reserve copied into more than one global owner;
4. a PSD carry column promoted to a positive arithmetic node source;
5. the Julia trace used as though its Mellin abscissa were zero;
6. a finite replay represented as a proof of GABPT;
7. a claim that this paper proves RH unconditionally.

10 Conclusion

The fixed-depth no-go is not the end of the parity-Hall programme. Adaptive depth proves positivity in the homogeneous product, and complete monotonicity of the critical ramp proves every smooth-interior history positive. The parity cocycle admits an exact PSD Julia implementation. These results compress the open producer to one globally owned activation-boundary scalar. That scalar is the correct target for the next proof or exact separator. GABPT remains open, and the Riemann Hypothesis remains unproved.